

CSR322:NATURAL LANGUAGE PROCESSING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Understand the evolution of natural language processing from classical methods to modern transformer-based approaches.

CO2 :: Explain why prompting works by analysing probabilistic language modelling and context driven reasoning.

CO3 :: Design effective, model-agnostic prompts using systematic prompt engineering methodologies.

CO4 :: Evaluate prompt outputs using structured evaluation frameworks and identify failure modes such as hallucination and prompt injection.

CO5 :: Design Retrieval-Augmented Generation systems as a standard pattern for grounding LLM outputs.

CO6 :: Apply production-level considerations such as cost, latency, scalability and reliability in LLM-based systems.

Unit I

Introduction to NLP : History and evolution of NLP, Key NLP tasks such as tokenization and tagging and parsing, Classification and generation, Challenges in processing natural language including ambiguity and context dependence, Overview of rule-based, Statistical and neural NLP paradigms, Transition from traditional NLP pipelines to deep learning and Transformer-based systems, Introduction to modern NLP libraries and frameworks

Unit II

Word Level : Corpora and their construction, Regular expressions, Finite state automata, Morphological parsing, Spelling error detection and correction, Words and word classes, Part-of-speech tagging

Syntactic Analysis : Context-free grammar, Constituency parsing, Dependency parsing, Probabilistic parsing

Semantic Analysis : Meaning representation, Ambiguity resolution, Word sense disambiguation

Unit III

Text Classification, Clustering and Classical NLP Context : Overview of text classification and clustering as NLP tasks, High-level understanding of supervised and unsupervised approaches, Conceptual role of feature-based models, Introduction to discourse processing including coherence and reference resolution, Fundamentals of natural language generation and its applications such as summarization and content creation, Conceptual overview of classical NLP models including n-gram language models, Hidden Markov models and context-free grammars, Limitations and their role in the evolution toward neural and transformer-based NLP systems

Unit IV

Transformers and Hugging Face Workflows : Transformer architecture and self-attention mechanisms, Encoder-only, Decoder-only and encoder-decoder architectures, Overview of pretrained transformer models such as BERT, RoBERTa, GPT, T5 and similar architectures, Transfer learning and fine-tuning strategies, Hugging Face ecosystem including datasets, Tokenizers and transformers, Building end-to-end NLP pipelines using transformer workflows

Unit V

Transformers : Introduction to transformer architecture, Self-attention mechanism, Encoder-Decoder structure, Overview of BERT, GPT, T5, ALBERT, Fine-tuning pre-trained transformers for specific tasks, Classification, Summarization, Question answering

Transfer Learning : Introduction to transfer learning, Implementing transformer models using libraries like Hugging Face Transformers

Unit VI

Applications and Building Models : Text Classification: Techniques and algorithms, Named Entity Recognition (NER): Applications in information extraction, Question Answering Systems: Applications in chatbots and virtual assistants, Text Summarization: Techniques and models, Machine Translation: Applications in global communication, Sentiment Analysis: Applications in social media monitoring and market analysis.

List of Practicals / Experiments:

list of practicals

- Explore and compare different tokenization techniques including BPE and Sentence Piece
- Use Hugging Face tokenizers and transformer models for basic NLP tasks
- Perform zero-shot and few-shot text classification using prompt-based approaches
- Build a text summarization pipeline using a pre-trained transformer
- Compare prompt-based NLP solutions with fine-tuned transformer models
- Evaluate NLP models using standard metrics and perform error analysis
- Analyze the impact of tokenization and prompt design on model outputs
- Develop an NLP pipeline combining tokenization, modelling, and evaluation

Text Books:

1. MASTERING LARGE LANGUAGE MODELS: ADVANCED TECHNIQUES, APPLICATIONS, CUTTING-EDGE METHODS, AND TOP LLMS by SANKET SUBHASH KHANDARE, BPB Publication, New Delhi
2. BUILDING TRANSFORMER MODELS WITH PYTORCH 2.0 by PREM TIMSINA, BPB PUBLICATIONS
3. MASTERING LARGE LANGUAGE MODELS by SANKET SUBHASH KHANDARE, BPB PUBLICATIONS

References:

1. PRACTICAL NATURAL LANGUAGE PROCESSING A COMPREHENSIVE GUIDE TO BUILDING REAL-WORLD NLP SYSTEMS by SOWMYA VAJJALA, BODHISATTWA MAJUMDER, ANUJ GUPTA, HARSHIT SURANA, SHROFF/O'REILLY